DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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LAB REPORT

Computer Networks Lab

Cisco Packet Tracer experiments: LAN connectivity and router configuration

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EXPERIMENT 1

Introduction to Packet Tracer, Peer-to-Peer Communication, Study of Cables and its Color Codes

Date
23-10-2026

AIM

To study Cisco Packet Tracer, connect two PCs through a switch and check the communication between them using ping, and to study the straight-through and crossover cables and their color codes.

SOFTWARE USED

Cisco Packet Tracer

THEORY

Cisco Packet Tracer is a network simulation software made by Cisco. We can create networks, configure devices and test them without using real hardware. In peer-to-peer communication the computers talk to each other directly without any server. A switch is used to connect devices of the same network and it forwards data using MAC address.

Ethernet cables have 8 wires (4 twisted pairs) and RJ-45 connectors are used at the ends. There are two wiring standards, T568A and T568B.

Wire color order

Pin	T568A	T568B
1	White-Green	White-Orange
2	Green	Orange
3	White-Orange	White-Green
4	Blue	Blue
5	White-Blue	White-Blue
6	Orange	Green
7	White-Brown	White-Brown
8	Brown	Brown

Cable types

Straight-through

Wiring
Same standard on both ends (T568B – T568B)

Connects
Different devices: PC to switch, switch to router

Crossover

Wiring
T568A on one end and T568B on the other end

Connects
Similar devices: PC to PC, switch to switch, router to router

IP ADDRESSES USED

Device	IP address	Subnet mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0

PROCEDURE

1. Open Cisco Packet Tracer.
2. Take two PCs (PC0 and PC1) from End Devices and one 2960 switch from Switches and place them in the workspace.
3. Select Copper Straight-Through cable and connect PC0 to Fa0/1 of the switch and PC1 to Fa0/2 of the switch.
4. Click on PC0 → Desktop → IP Configuration and give IP address 192.168.1.1 and subnet mask 255.255.255.0.
5. Click on PC1 → Desktop → IP Configuration and give IP address 192.168.1.2 and subnet mask 255.255.255.0.
6. Wait till all the link lights become green.
7. Open PC0 → Desktop → Command Prompt and type `ping 192.168.1.2`
8. Check the replies.

OUTPUT

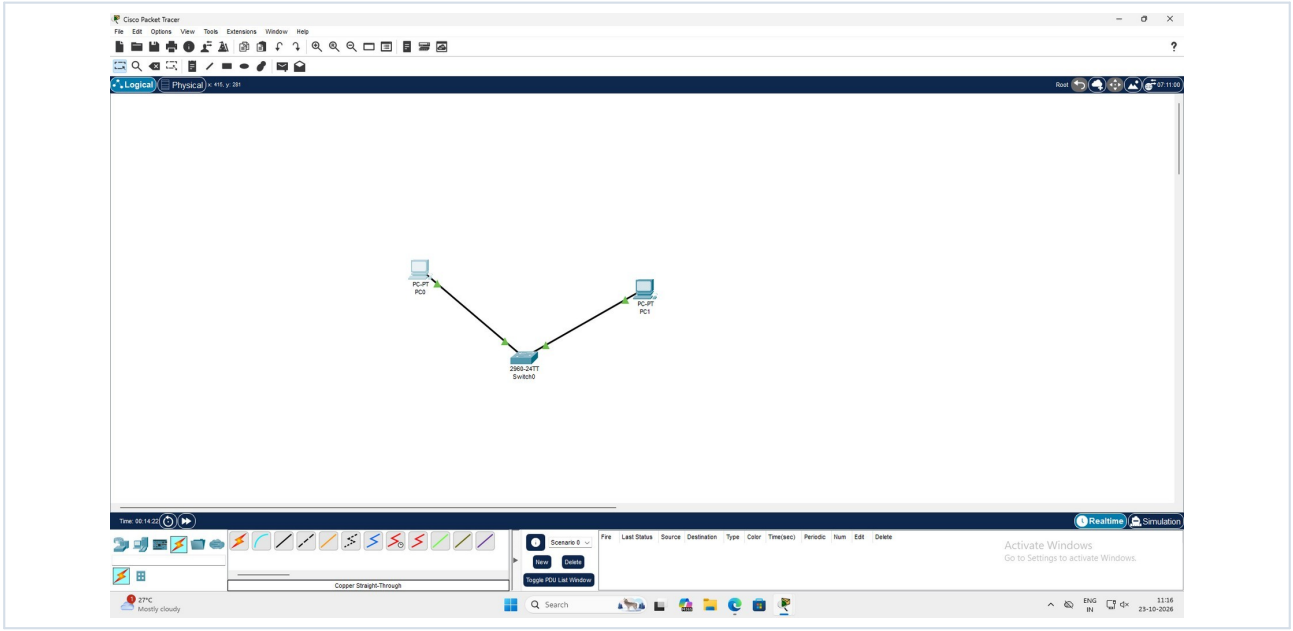


Fig. 1.1 – Topology: PC0 and PC1 connected to Switch0 with straight-through cables

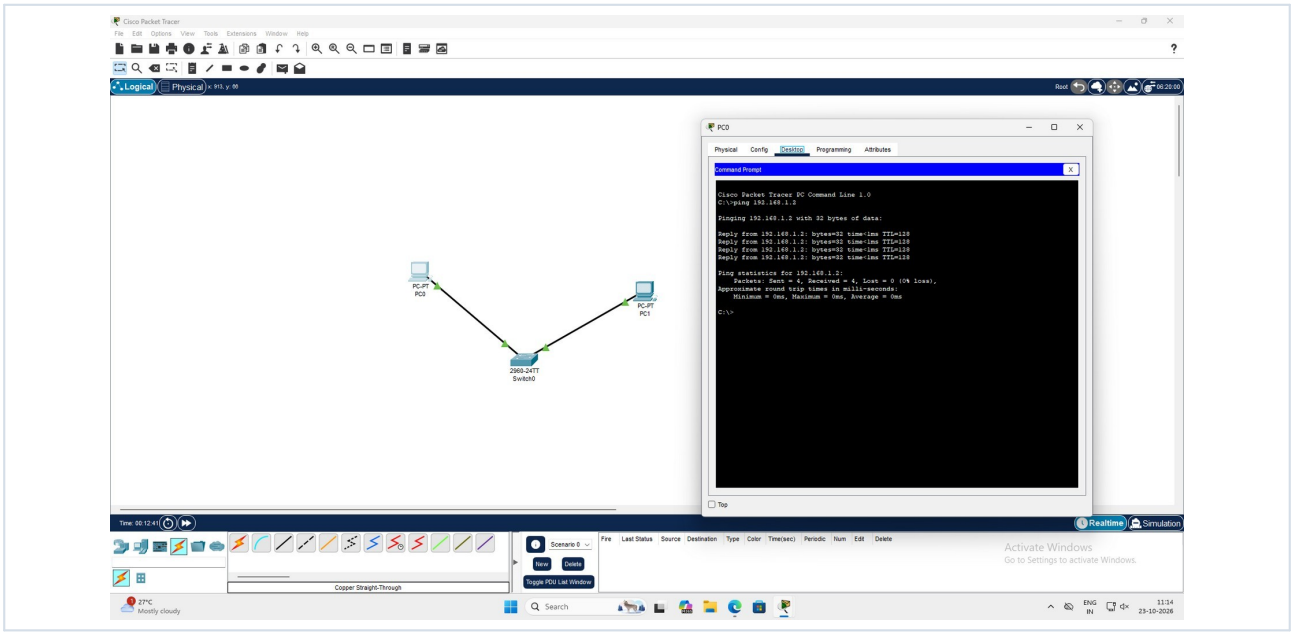


Fig. 1.2 – Ping from PC0 to PC1 (4 sent, 4 received, 0% loss)

RESULT

Thus the two PCs were connected using a switch in Cisco Packet Tracer and the ping from PC0 to PC1 was successful (4 sent, 4 received, 0% loss). The straight-through and crossover cables and their color codes were studied.

EXPERIMENT 2

Router Configuration (Creating Passwords, Configuring Interfaces)

Date
23-10-2026

AIM

To configure a router by setting passwords and IP addresses on its interfaces, and to check the communication between two PCs in different networks through the router.

SOFTWARE USED

Cisco Packet Tracer

THEORY

A router is used to connect two or more different networks. It forwards packets based on IP address. Each interface of the router is in a different network and it works as the default gateway for the PCs in that network. Passwords are given to the router for security – console password for console access, enable password for privileged mode and VTY password for remote (Telnet) access. By default router interfaces are shut down, so we have to use the `no shutdown` command to turn them on.

IP ADDRESSES USED

Device	Interface	IP address / mask	Default gateway
Router (R1)	Gig0/0	192.168.1.1 / 255.255.255.0	–
Router (R1)	Gig0/1	192.168.2.1 / 255.255.255.0	–
PC2	NIC	192.168.1.2 / 255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.2 / 255.255.255.0	192.168.2.1

PROCEDURE

1. Open Cisco Packet Tracer and place one 2911 router and two PCs (PC2 and PC3).
2. Connect PC2 to Gig0/0 and PC3 to Gig0/1 of the router using copper cables.
3. Click on the router and go to CLI tab. Type `no` for the initial configuration dialog and press Enter.
4. Type the following commands to set the passwords:

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#enable password cisco
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#exit
```

5. Type the following commands to configure the interfaces:

```
R1(config)#interface gig0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface gig0/1
R1(config-if)#ip address 192.168.2.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#end
R1#show ip interface brief
```

6. In PC2 give IP 192.168.1.2, subnet mask 255.255.255.0 and default gateway 192.168.1.1.
7. In PC3 give IP 192.168.2.2, subnet mask 255.255.255.0 and default gateway 192.168.2.1.
8. Open Command Prompt in PC2 and type `ping 192.168.2.2`

OUTPUT

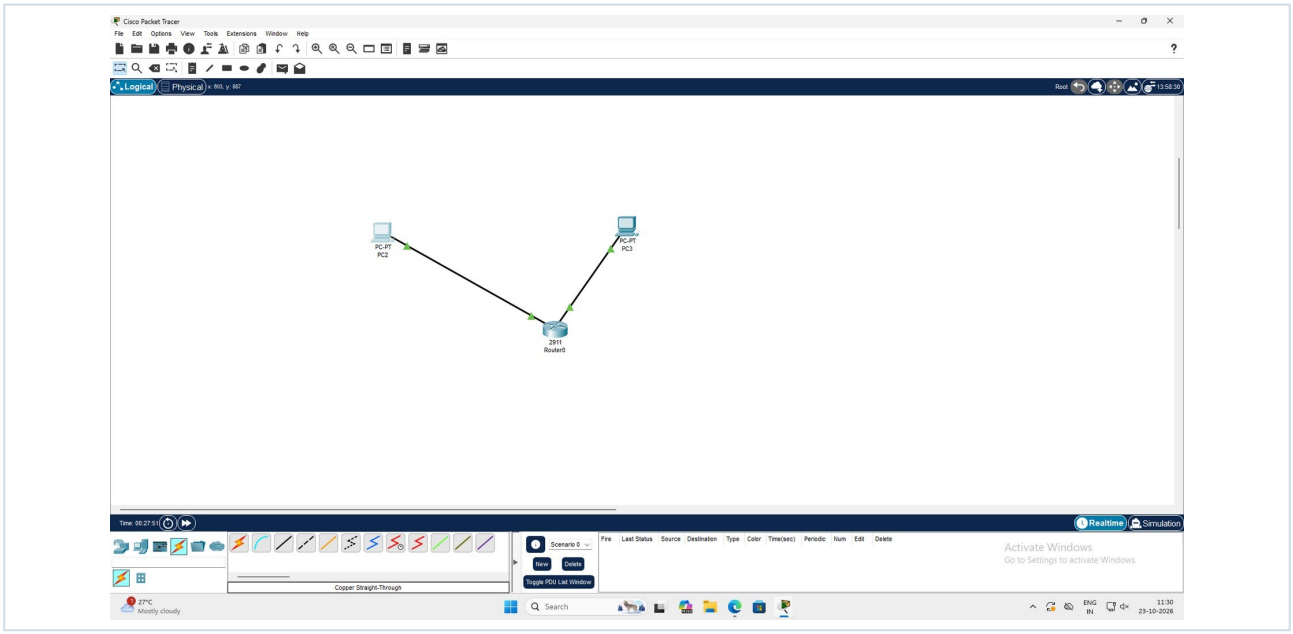


Fig. 2.1 – Topology: PC2 and PC3 connected to the 2911 router (R1)

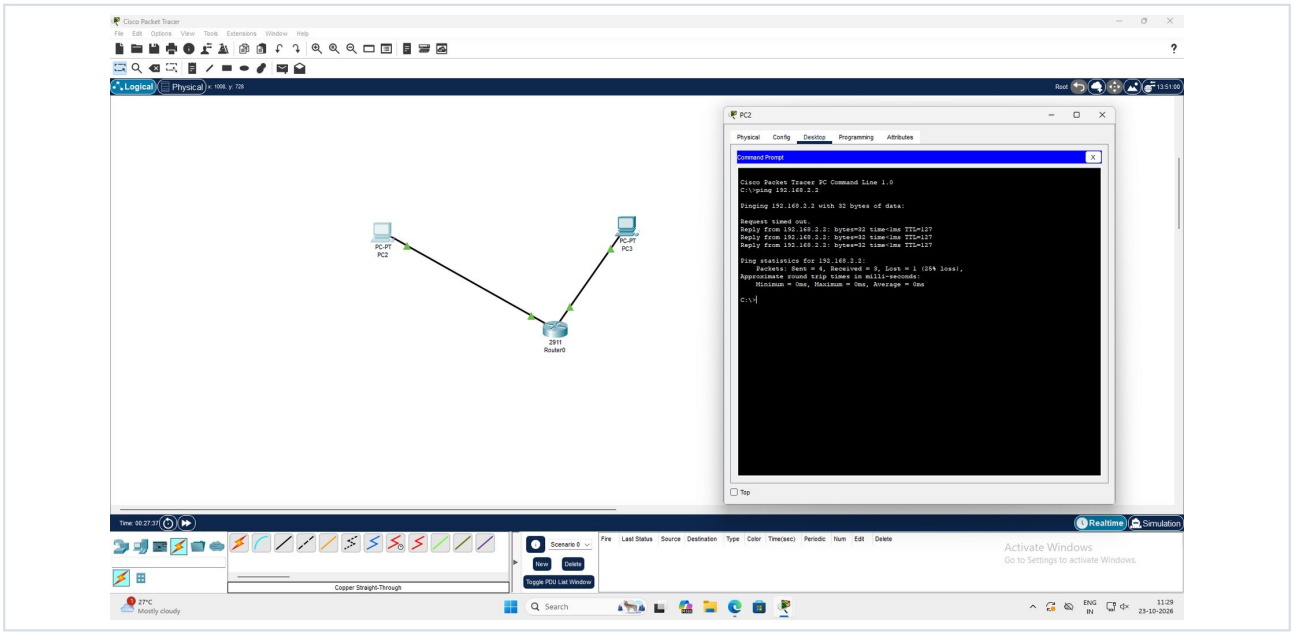


Fig. 2.2 – Ping from PC2 to PC3 through the router

The first request timed out because of ARP, after that all replies came.

RESULT

Thus the router was configured with passwords and IP addresses on its interfaces, and the two PCs in different networks were able to communicate through the router.